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RESEARCH ARTICLE

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Can proactive boundary-spanning search enhance green innovation? The mediating role of organizational resilience

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Abstract

Green innovation is critical for firms' sustainable development, and its antecedents have attracted extensive attention. However, how proactive boundary-spanning search enhances firms' green innovation remains unclear. Building on the resource-based view and dynamic capabilities theory, we explore the impact of proactive boundary-spanning search on firms' green innovation and investigate the mediating role of two types of organizational resilience. Survey data are collected from 218 manufacturing firms in China, and hierarchical regression and bootstrap analysis are used to test hypotheses. The results demonstrate that proactive boundary-spanning search *not only has an* inverted U-shaped relationship with radical green innovation and incremental green innovation, but also has an inverted U-shaped relationship with precursor resilience and improvisation resilience. Moreover, the results show that precursor resilience and improvisation resilience partially mediate the inverted U-shaped relationship between proactive boundary-spanning search and the two types of green innovation. This research reveals the nonlinear relationship between proactive boundary-spanning search and green innovation, which enriches the research of green innovation by considering organizational resilience.

KEYWORDS
green innovation, organizational resilience, proactive boundary-spanning search, sustainable development

1 | INTRODUCTION

In recent years, firms pay increasing attention to the impact of their decision-making and business behavior on the environment and conduct green innovation to pursue long-term and sustainable development (Melander, 2018; Shahzad et al., 2021; Soewarno et al., 2019). Green innovation is the process of promoting and improving products or production in energy conservation, pollution prevention, waste recycling, green product design, and environmental management (Chang, 2011; Chen, 2008; Martínez-Ros & Kunapatarawong, 2019), which is a win-win solution for firms to reduce the conflict between economic development and environmental protection (Bhatia, 2021; Hao et al., 2022). On the one hand, green innovation can help firms realize environmental benefits, such as saving energy, reducing carbon dioxide emission, and improving product recovery rate (Zhang et al., 2020); on the other hand, green innovation is beneficial for firms to increase economic benefits by promoting green production capacity, improving firms' image, and gaining customer recognition (Huang et al., 2022; Wang, 2020). Therefore, green innovation is an important strategy for firms to achieve sustainable development.

Green innovation is characterized by technological complexity, high cost, and high risk (De Marchi, 2012), which requires more knowledge resources than traditional innovation (Martínez-Ros & Kunapatarawong, 2019; Pacheco et al., 2018), especially the knowledge and skills of key technologies needed to eliminate environmental pollution (Liao & Tsai, 2019; Sahoo et al., 2022). However, due to the lack of internal knowledge resources, firms must search external knowledge to effectively explore green technologies (Du et al., 2018;

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Song et al., 2020). As an effective way to obtain external knowledge, proactive boundary-spanning search is the process for firms to search and identify external knowledge that has not been widely accepted and has potential value ahead of competitors (Yang et al., 2021). Proactive boundary-spanning search facilitates firms to overcome organizational rigidity and capability traps, and it is an effective way for firms to break through the bottleneck due to lack of internal knowledge and promote innovation (Zhang et al., 2017). Existing studies have found that many factors affect firms' green innovation, such as market demand (Mondéjar-Jiménez et al., 2015), institutional pressure (Liao, 2018; Lin & Ho, 2016), environmental regulation (Zhang et al., 2018), stakeholder pressure (Kawai et al., 2018; Song et al., 2020), external knowledge resources (Flor et al., 2018; Martínez-Ros & Kunapatarawong, 2019; Mothe & Nguyen-Thi, 2017; Wang et al., 2020), and big data and environmental ethics (El-Kassar & Singh, 2019). However, there are few studies on the relationship between proactive boundary-spanning search and green innovation. Resource-based view (RBV) suggests that valuable, rare, and inimitable resources determine the sustainable competitive advantage of firms. Firms need to obtain knowledge from different external organizations to facilitate green innovation. Moreover, under the new normal of volatility, uncertainty, complexity, and ambiguity (VUCA), proactive boundary-spanning search can help firms gain heterogeneous knowledge ahead of competitors and better adapt to the complex and unpredictable external environment. In this way, firms will have more opportunities to effectively carry out green innovation (De Marchi & Grandinetti, 2013; Zhao et al., 2018) and maintain first-mover green advantage. Therefore, it is worth exploring how proactive boundary-spanning search affects green innovation.

In the VUCA environment, proactive boundary-spanning search alone cannot effectively promote green innovation. It is important for firms to build organizational resilience to cope with risks facing green innovation. Organizational resilience, a hot topic and an emerging field of management research (Andersson et al., 2019; Carmeli et al., 2020), refers to the ability to anticipate potential threats, effectively deal with adverse and unexpected events (such as COVID-19 pandemic and natural disasters), and adjust to environmental changes (Annarelli & Nonino, 2016; Jia et al., 2020; Linnenluecke, 2017; Ortiz-de-Mandojana & Bansal, 2016). Drawing on dynamic capabilities theory, organizational resilience can be regarded as a dynamic capability (Blessley & Mudambi, 2022; Do et al., 2022; Silva & Ruel, 2022) that firms deal with shocks and unexpected threats in complex and uncertain environments (Duchek, 2020; Linnenluecke, 2017; Sajko et al., 2021). Thus, organizational resilience enables firms to rapidly identify and respond to emergencies and recover or even rebound in adversity (Audretsch & Belitski, 2021; Groenendaal & Helsloot, 2020; Williams & Shepherd, 2016), thereby effectively coping with various risks and crises in the process of green innovation. Previous studies mainly focused on the antecedents of organizational resilience, including organizational learning (Koronis & Ponis, 2018; Linnenluecke, 2017), social capital (Jia et al., 2020), human resource management (Andersson et al., 2019; Bouaziz & Hachicha, 2018), and external knowledge (Alonso et al., 2020; Carmeli et al., 2020). Some current research also paid attention to the consequences of organizational resilience and explored the impact of organizational resilience on innovation (Akgün & Keskin, 2014; Athota & Malik, 2019; Fandiño et al., 2019) or performance (Gölgeci & Kuivalainen, 2020; Melián-Alzola et al., 2020). Although current research has emphasized that resilience is conducive to innovation (Akgün & Keskin, 2014; Chowdhury et al., 2019; Do et al., 2022; Duchek, 2020), it is still unclear how firms can improve organizational resilience through boundary-spanning search to deal with risks in the process of green innovation. Therefore, it is necessary to explore the role of organizational resilience in proactive boundary-spanning search and green innovation.

To fill the research gaps, based on the RBV and dynamic capabilities theory, we take Chinese manufacturing firms as the sample and explore the impact of proactive boundary-spanning search and organizational resilience on green innovation. The aims of this research are to uncover whether and how proactive boundary-spanning search affects green innovation and examine the mediating role of organizational resilience. First, by considering two types of green innovation, we propose that proactive boundary-spanning search exerts inverted U-shaped effects on radical and incremental green innovations. Second, we divide organizational resilience into precursor resilience and improvisation resilience and argue that there are inverted U-shaped relationships between proactive boundary-spanning search and both types of resilience. Third, we regard organizational resilience as mediator and examine its mediating role between proactive boundary-spanning search and green innovation. The rest of this paper is structured as follows. Section 2 presents the research hypotheses. Section 3 reports the data collection and research design. Section 4 shows the data processing and results. Section 5 concludes the paper with discussions on research implications and limitations. Section 6 presents the conclusions.

2 | RESEARCH HYPOTHESES

Previous research has suggested that RBV can be used to explain how firms obtain knowledge from different external organizations to facilitate green innovation (Serrano-García et al., 2021; Tariq et al., 2017). Dynamic capabilities theory is used to describe how firms anticipate and cope with unexpected events to build resilience by acquiring external knowledge (Do et al., 2022; Haarhaus & Liening, 2020; Teece, 2014) and further achieving green innovation. Thus, drawing on RBV and dynamic capabilities theory, this study considers that proactive boundary-spanning search affects green innovation, and organizational resilience plays a mediating role between proactive boundary-spanning search and green innovation. The theoretical model of this study is shown in Figure 1.

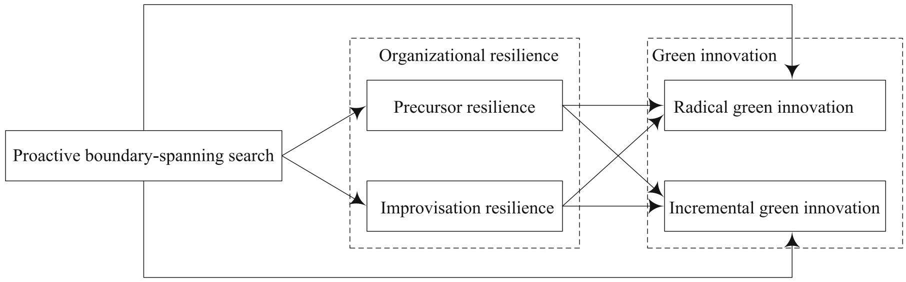

FIGURE 1 Theoretical model

2.1 | Proactive boundary-spanning search and green innovation

Green innovation can be radical or incremental (Hayaeian et al., 2022; Subramaniam & Youndt, 2005; Liao & Zhang, 2020). Radical green innovation is to design and develop new environment-friendly products to enter new green markets or emerging markets (Cui & Wang, 2022), while incremental green innovation aims to improve firms' existing environment-friendly products to extend the existing market (Flor et al., 2018; Liu, 2021). In the new normal of VUCA, green innovation will face greater risks, which requires firms to predict environmental changes in advance and adopt proactive behaviors (Bhatia, 2021). As a typical proactive behavior, proactive boundary-spanning search not only helps firms grasp the policies related to green development (De Marchi, 2012) and the development trend of green technology (Demirel & kesidou, 2019), but also provides firms with more opportunities to identify the green needs of customers and other stakeholders (Ben Arfi et al., 2018). In this regard, proactive boundary-spanning search is critical for both radical and incremental green innovations.

According to RBV, proactive boundary-spanning search is an important channel for firms to obtain external knowledge, which is critical to facilitate radical innovation (Flor et al., 2018; Hayaeian et al., 2022; Jin & Shao, 2022). Therefore, we propose that proactive boundary-spanning search may have a positive impact on firms' radical green innovation. First, proactive boundary-spanning search can help firms overcome their own development bottlenecks, obtain advanced green technologies (Hu et al., 2021), green materials, and unique knowledge (Liu, 2021; Melander, 2018). In this way, firms can constantly generate breakthrough ideas (Gao et al., 2022; Zhang & Hu, 2017) and then enter new markets to reap benefits from radical green innovation (Zhang et al., 2020). Second, proactive boundary-spanning search is conducive to obtaining cutting-edge technologies of environmental protection and ecological practice from external organizations (Cruz-González et al., 2014; Ma et al., 2018), and extracting novel knowledge in ecological design, environmental protection materials, cleaner production and green packaging. Thus, radical green innovation will be fostered. Third, by conducting proactive boundary-spanning search, firms can identify the government's green subsidies, market access, emission standards and relevant laws and regulations in advance (Hoppmann et al., 2021) and improve the sensitivity of their green activities, thereby realizing radical green innovation.

However, excessive proactive boundary-spanning search may be detrimental to green innovation. First, it requires firms to have stronger abilities to identify, absorb and integrate knowledge (Li et al., 2019; Song et al., 2020). If there is too much knowledge and firms' absorptive capability is insufficient, it will lead to knowledge redundancy and overload (Shi et al., 2019), which will increase the difficulty of absorbing and utilizing new knowledge (Martínez-Ros & Kunapatarawong, 2019), thus impeding radical green innovation. Second, excessive information and knowledge can easily cause attention allocation issues (Ardito & Petruzzelli, 2017), and thus, firms may ignore the integration of heterogeneous knowledge. In this vein, firms may not be able to apply new knowledge to deal with changes in the external environment, which will hinder the development of more environment-friendly technologies and products. Therefore, excessive proactive boundary-spanning search may have a negative impact on radical green innovation.

Taken together, we argue that proactive boundary-spanning search may play a "double-edged sword" role in firms' radical green innovation. Specifically, when the level of proactive boundary-spanning search is moderate, it will benefit radical green innovation. However, when the level of proactive boundary-spanning search is too low, the knowledge may be insufficient to facilitate radical green innovation, and when the level of proactive boundary-spanning search is too high, excessive search will increase knowledge redundancy, thereby hindering radical green innovation. Hence, we posit:

> **H1a.** Proactive boundary-spanning search has an inverted U-shaped impact on radical green innovation.

Similarly, proactive boundary-spanning search may have a curvilinear impact on firms' incremental green innovation. On the one hand, firms can obtain process improvement methods and technologies of green product design through proactive boundary-spanning search, identify potential technology and product needs, and enrich their own knowledge sources (Xie et al., 2022; Zhang et al., 2022). In this way, green and environment-friendly products can be designed, thereby promoting incremental green innovation. On the other hand, proactive boundary-spanning search emphasizes that firms ahead of competitors pay more attention to the potential needs of existing customers and scan green practices and new environmental knowledge of existing markets (Yu et al., 2020), which helps firms improve the restrictions of existing businesses and product series (Liu, 2021).

Accordingly, existing green products can be improved by orchestrating new knowledge and existing knowledge, and then incremental green innovation is facilitated.

However, excessive proactive boundary-spanning search may be harmful to incremental green innovation. First, knowledge acquired by excessive proactive boundary-spanning search may be too complex and novel and distant from existing knowledge (Flor et al., 2018; Yu et al., 2020). As a result, the acquired knowledge cannot be effectively evaluated and handled, which will make it difficult for firms to correctly understand and use new knowledge to improve existing products (Díaz-Díaz & de Saá-Pérez, 2014), thus hampering incremental green innovation. Second, excessive proactive boundary-spanning search will consume a lot of resources, and firms need to invest more human and material resources to identify, screen, and utilize the new acquired knowledge. Therefore, excessive knowledge search will increase the costs of firms (De Marchi, 2012; Flor et al., 2018) and lead to the high burden of knowledge management (Camisón et al., 2018; Faems et al., 2010; Shi et al., 2019) and coordination (Faems et al., 2010; Ferreras-Méndez et al., 2016; Leiponen & Helfat, 2010). Consequently, the cost of absorbing and utilizing external knowledge may exceed the benefits of improving green products (Flor et al., 2018), thereby hindering incremental green innovation.

In sum, when the level of proactive boundary-spanning search is moderate, it will improve incremental green innovation. However, when the level of proactive boundary-spanning search is too low or too high, it will hamper the realization of incremental green innovation and has an adverse impact on incremental green innovation. Hence, we posit:

> **H1b.** Proactive boundary-spanning search has an inverted U-shaped impact on incremental green innovation.

2.2 | Proactive boundary-spanning search and organizational resilience

Organizational resilience is the ability of an organization to predict potential threats, effectively deal with unexpected events, and adapt to environmental changes (Annarelli & Nonino, 2016; Linnenluecke, 2017; Ortiz-de-Mandojana & Bansal, 2016). Organizations should not only take proactive actions to predict the unexpected events and potential crises they may face in the future, but also take improvisational actions to deal with the unexpected events (Duchek, 2020; Jia et al., 2020; Koronis & Ponis, 2018). Based on this, from anticipation and improvisation perspectives, this study divides organizational resilience into precursor resilience and improvisation resilience. Precursor resilience is the ability that firms predict and prevent unexpected events before disruptions (Blessley & Mudambi, 2022; Boin & van Eeten, 2013), while improvisation resilience is the ability that firms improvise to deal with and recover from the events after disruptions (Duchek, 2020; Jia et al., 2020). Studies have suggested that firms can have both resilience at the same time (Ates & Bititci, 2011; Jia et al., 2020).

Specifically, before the unexpected events, the new and diversified knowledge obtained by proactive boundary-spanning search enables firms to quickly anticipate and predict potential unexpected events (Duchek, 2020; Ji et al., 2020), such as customers' latent green needs (Kong et al., 2020), upcoming environmental issues, and environmental market trends (Bhatia, 2021). Firms can then envisage a variety of ways to deal with these unexpected events and seek the most appropriate solutions in advance (Duchek, 2020), thereby enhancing precursor resilience. In addition, external knowledge acquired by proactive boundary-spanning search enhances firms' ability to cultivate external awareness (Barasa et al., 2018) and predict latent emergencies in advance (Hillmann, 2021), thus building precursor resilience. However, excessive proactive boundary-spanning search will lead to knowledge redundancy, and it will be difficult for firms to identify the most valuable information from redundant knowledge (Ferreras-Méndez et al., 2016), which will narrow firms' attention (Hoppmann et al., 2021) and then hamper firms' ability to predict and handle potential emergencies. Moreover, too little proactive boundary-spanning search cannot enable firms to accurately identify and predict the trend of environmental changes, which further makes it difficult for firms to take proactive actions for unexpected events, and then impede precursor resilience. Taken together, when proactive boundary-spanning search is moderate, it will improve precursor resilience. In contrast, when proactive boundary-spanning search is too little or too much, it will reduce precursor resilience. Hence, we propose:

> **H2a.** Proactive boundary-spanning search has an inverted U-shaped effect on precursor resilience.

Proactive boundary-spanning search can facilitate firms to enhance market insights (Do et al., 2022), and assist firms to identify and detect knowledge associated with current threats such as abnormal information of customers, green specifications and environmental standards of suppliers (Ji et al., 2020; Shah & Soomro, 2021), new skills of competitors, and new environmental regulations of government. In this vein, firms can identify environmental requirements and seize new opportunities in the external environment to deal with the crises, thus enhancing improvisation resilience (Tengblad & Oudhuis, 2018). Moreover, the valuable knowledge identified by proactive boundary-spanning search is critical to overcoming the challenges of emergencies, as emergencies highly rely on the availability of knowledge (Brandon-Jones et al., 2014). In this vein, firms can take real-time improvisational actions to respond to environmental uncertainty (Liao, 2018; Zhang et al., 2017) and emergent situations. Thus, firms with proactive boundary-spanning search will have stronger improvisation resilience. However, excessive proactive boundary-spanning search is not conducive to firms' timely response to environmental changes and limits firms' improvisation in dealing with emergencies and rapid recovery, thus damaging improvisation resilience. Similarly, little proactive boundary-spanning search will cause knowledge deficit, and firms cannot grasp the external environmental change and cope with unexpected events immediately, which reduces improvisation resilience. In sum, when the level of proactive boundary-spanning search is moderate, it will enable firms to build improvisation resilience; otherwise, it will hamper improvisation resilience. Hence, we propose:

H2b. Proactive boundary-spanning search has an inverted U-shaped effect on improvisation resilience.

2.3 | The mediating role of organizational resilience

Organizational resilience not only helps firms recover from adversity but also facilitates firms to create new opportunities to achieve leapfrog development. Therefore, organizational resilience plays an important role in firms' green innovation and has become the driving factor of green innovation and the source of competitive advantage (Teixeira & Werther, 2013). Since moderate proactive boundary-spanning search is conducive to improving organizational resilience, which in turn promotes firms' green innovation, this study proposes that organizational resilience plays a mediating role between proactive boundary-spanning search and firms' green innovation.

Specifically, if the level of proactive boundary-spanning search is moderate, diverse knowledge obtained by proactive boundary-spanning search can increase firms' sensitivity to the environment (Audretsch & Belitski, 2021), enhance firms' predictive capability, and thus foster firms' precursor resilience. Precursor resilience can guide firms to effectively explore potential risks (Duchek, 2020), and take proactive steps to buffer the negative impact by reconfiguring external and internal knowledge (Parker & Ameen, 2018). In this way, firms can seize unexpected opportunities, create novel ideas, and develop disruptive green products matching the changing environment, so as to further promote radical green innovation. Moreover, firms with precursor resilience are predictive and can detect the potential needs of customers and track changes in industry innovation (Teixeira & Werther, 2013; Yao et al., 2021). Accordingly, firms can create breakthrough green products and services to satisfy the need of emerging market, thereby facilitating radical green innovation. Similarly, firms with precursor resilience not only can detect potential changes in the external environment (Wu & Wu, 2014), but also can effectively develop new solutions to deal with unexpected events and uncertainties caused by changes (Do et al., 2022). As a consequence, firms can timely adjust and utilize the acquired knowledge, promptly develop solutions to improve existing green products and processes, and then realize incremental green innovation. However, excessively low or excessively high proactive boundary-spanning search will provide insufficient or redundant knowledge, which may hamper precursor resilience. If precursor resilience of firms is weak, radical green innovation and incremental green innovation will be hindered. Taken together, moderate proactive boundary-spanning search facilitates precursor resilience, which can allow firms to predict unexpected changes and improve their environmental adaptability and market competitiveness, thereby enhancing radical and incremental green innovations. Thus, we propose the following:

H3a. Precursor resilience mediates the relationship between proactive boundary-spanning search and radical green innovation.

H3b. Precursor resilience mediates the relationship between proactive boundary-spanning search and incremental green innovation.

By the same token, moderate proactive boundary-spanning search can help firms acquire market knowledge and technical knowledge ahead of competitors, and diversified knowledge facilitates firms to respond to customer needs and changes in the external environment in time (Ji et al., 2020; Liao & Tsai, 2019), thus building firms' improvisation resilience. On the one hand, when firms with improvisation resilience face unexpected events, they have stronger crisis response ability and faster response speed. Firms can not only identify crisis events and develop ad hoc solutions to better adapt to the impact of emergencies (Kendra & Wachtendorf, 2003), but also put forward new ideas to deal with acute crisis (Athota & Malik, 2019; Do et al., 2022), thereby promoting the development of new green products and achieving radical green innovation. On the other hand, when firms have improvisation resilience, even the subtle information threatening firms in a complex environment will be captured in time, and then be analyzed to curb the spread of the crisis (Bode & Macdonald, 2017). Accordingly, firms can rapidly respond to and easily identify the opportunities from disruptions by reconfiguring knowledge (Do et al., 2022), and adjust and update existing green products and processes, so as to realize incremental green innovation. In contrast, when the level of proactive boundary-spanning search is too low or too high, improvisation resilience will be hindered, and it becomes difficult for firms to deal with unexpected events, thus impeding radical green innovation and incremental green innovation. In sum, moderate proactive boundary-spanning search helps firms improve improvisation resilience, which further facilitates firms to enhance their ability to deal with emergencies and then promotes radical and incremental green innovations. Therefore, we develop the following:

H3c. Improvisation resilience mediates the relationship between proactive boundary-spanning search and radical green innovation.

H3d. Improvisation resilience mediates the relationship between proactive boundary-spanning search and incremental green innovation.

3 | METHOD

3.1 | Data collection

The manufacturing industry is the main body of China's national economy and is the foundation for China's consolidation and development. Hence, manufacturing innovation is not only critical to advance technologies but also an important way to promote high-quality development. However, manufacturing firms have the greatest negative impact on the environment (Zhang et al., 2020), leading to serious environmental problems, such as excessive carbon emissions. Therefore, it is urgent for manufacturing firms to carry out green innovation. As such, we chose the manufacturing industry as the research context and collected data from manufacturing firms in six major cities in China: Shenzhen, Suzhou, Guangzhou, Beijing, Shanghai, and Xi'an. Shenzhen and Guangzhou are located in southern China, with high level of economic development. Shanghai and Suzhou, located in eastern China, are developing rapidly. Located in northern China, Beijing is the capital with an average level of economic development. Xi'an is a city with a low level of economic development in western China. These geographically diverse cities have manufacturing firms with different levels of green innovation. Thus, these six cities can reflect the degree of China's economic and manufacturing development.

The respondents are middle and senior managers of manufacturing firms as they are familiar with the firms' development status and future strategy. From April to June 2021, 350 questionnaires were distributed in three ways. First, we visited 48 manufacturing firms in the six cities, interviewed some middle or senior managers of each firm, and asked them to fill out the paper questionnaire. A total of 48 questionnaires were distributed, all of which were returned. Second, we selected students working in manufacturing firms in our MBA and EMBA classes and asked them to fill out the paper questionnaire. A total of 90 questionnaires were distributed, and 86 were completed. Third, we sent the electronic questionnaire to the middle and senior managers of manufacturing firms in the six cities by email. A total of 212 emails were sent. One week later, a reminder email was sent to the manager who did not reply, and 126 were finally returned. In total, 218 valid questionnaires were obtained after eliminating 42 incomplete responses, with an effective response rate of 62.29%. The sample distribution is shown in Table 1.

TABLE 1 Sample characteristics

Characteristics of firms	Number of firms	Percentage
Firm age		
5 or less	17	7.80%
6–10	32	14.68%
11–15	36	16.51%
16–20	45	20.64%
21 or more	88	40.37%
Number of employees		
500 or less	36	16.51%
501–1000	63	28.90%
1001–1050	26	11.93%
1051–2000	28	12.84%
2001or more	65	29.82%
Ownership		
State-owned firms	69	31.66%
Private firms	47	21.56%
Sino-foreign firms	51	23.39%
Foreign-invested firms	51	23.39%

3.2 | Measurement

All the scales in this study were adapted from existing scales. We adopted the method of translation and back-translation to ensure the accuracy of language expression. The questionnaire was designed in English and then was translated in Chinese. After discussion with three experts and five managers, unclear and ambiguous items were modified, and the final questionnaire was back-translated into English. Except for control variables, the items were measured by using 5-point Likert scales, with 1 *strongly disagree* to 5 *strongly agree*.

Proactive boundary-spanning search was adapted from Katila and Ahuja (2002) and Wang et al. (2020), including four items: "Our firm has many channels to search new domain knowledge ahead of competitors," "Our firm can search new domain knowledge about R&D, production, and marketing ahead of competitors," "Our firm can search new domain knowledge about technology and management ahead of competitors," and "Our firm can search more new knowledge ahead of competitors."

For organizational resilience, precursor resilience was adapted from Bode and Macdonald (2017) and Ji et al. (2020), including five items: "Our firm has an emergency plan to deal with unexpected events," "Our firm has the ability to deal with unexpected events," "Our firm has resources to deal with unexpected events," "Our firm can be alert to changes in the environment," and "Our firm can carry out early warning of unexpected events." Improvisation resilience was measured using five items adapted from Bode and Macdonald (2017) and Ji et al. (2020), including "Our firm can respond to unexpected events immediately," "Our firm can adapt to unexpected events," "Our firm can take quick action to deal with unexpected events," "Our firm can use resources to quickly solve unexpected events," and "Our firm can control unexpected events in time."

For green innovation, we divided it into radical green innovation and incremental green innovation. Radical green innovation was assessed using four items adapted from Flor et al. (2018) and Cui and Wang (2022), items are "Our firm introduces innovative ideas of new environment-friendly products or services," "Our firm develops new environment-friendly products or services," "Our firm masters new environment-friendly knowledge and skills in some fields," and "Our firm adopts new environment-friendly technologies to make major changes in products or services." Incremental green innovation was measured using four items adapted from Lennerts et al. (2020) and Liu (2021), items include "Our firm improves the quality of existing environment-friendly products or services," "Our firm improves the production efficiency of existing environment-friendly products or services," "Our firm consolidates the knowledge and skills of existing environment-friendly products or services," and "Our firm has the ability to improve existing environment-friendly products or services."

Previous studies have shown that firm age, firm scale, and firm ownership may affect green innovation. Specifically, most firms do not have the energy to consider social responsibility goals, such as environmental protection, until they have established a certain period of years (Kong et al., 2020; Liao, 2018). Therefore, the longer the firm is established, the easier it is for the firm to carry out green innovation. Research indicated that large-scale firms have rich financial and human capital and possess more technological opportunities and capabilities to engage in green innovation (Liao & Tsai, 2019; Martínez-Ros & Kunapatarawong, 2019; Moreno-Mondéjar et al., 2020; Triguero et al., 2013). Hence, large-scale firms are easier to conduct green innovation and need to be controlled. In addition, state-owned firms take the lead in undertaking social and economic responsibilities in China and will pay more attention to green innovation (Bai et al., 2019). Therefore, this study chose firm age, firm scale, and firm ownership as control variables. Among them, firm age was measured by the number of years since its establishment. Firm scale was measured by the number of employees. Taking foreign-invested firms as the reference group, firm ownership was measured by designing three dummy variables.

3.3 | Nonresponse bias and common method variance

To test nonresponse bias, we conducted *t*-test by comparing the top 50 respondents and the bottom 50 respondents (Dennehy et al., 2021). It was found that there were no significant differences in firm age ($t = .384$, $p = .702$), firm scale ($t = 1.018$, $p = .311$), and firm ownership ($t = .102$, $p = .919$) between these two groups. Therefore, there is no serious nonresponse bias in this study.

We employed three ways to test common method variance. First, when designing the questionnaire, we mainly referred to the mature scales in the current research, promised the respondents that there is no right or wrong answer, and collected data anonymously to reduce the impact of common method variance. Second, Harman's single-factor test was conducted, and exploratory factor analysis was used to analyze the principal components of all items; the first factor without rotation explained 35.595% of the total variance, which did not account for the majority and was less than 50% (Hair et al., 2016). Third, we made confirmatory factor analysis; compared with other models, the fit of one single-factor model ($\chi^2 = 2602.477$, $df = 209$, RMSEA = .230, TLI = .313, CFI = .378) is not good and is worse than other measurement models. Overall, there is no serious common method variance in this research.

3.4 | Reliability and validity analysis

We applied Cronbach's alpha and composite reliability (CR) to test reliability. As displayed in Table 2, the values of Cronbach's alpha and CR are bigger than the recommended value of .7 (Fornell & Larcker, 1981; Nunnally, 1978), suggesting good reliability.

We assessed construct validity, convergent validity, and discriminant validity. Construct validity was examined by performing exploratory factor analysis. Five factors were extracted according to the rule that the eigenvalue is greater than 1, and the factor loadings of all items in Table 2 are greater than .5, demonstrating good construct validity. Convergent validity was tested by computing average variance extracted (AVE) and confirmatory factor analysis. As shown in Table 2, the AVE values of all variables are in the range of .657 and .773, and each value is greater than the recommended level of .5. Thus, the measures have good convergent validity. Moreover, the results in Table 3 show that the five-factor model yields a reasonable fit ($\chi 2 = 473.597$, $df = 199$, RMSEA = .080, TLI = .917, and CFI = .929), demonstrating convergent validity. Regarding discriminant validity, as indicated in Table 3, the fit indices of the five-factor model are better than those of other models. In addition, the square root of AVE in Table 4 exceeds the correlation between the constructs, thus demonstrating good discriminant validity.

4 | RESULTS

4.1 | Descriptive statistics and correlation analysis

The variables' descriptive statistics and correlations are shown in Table 4. Proactive boundary-spanning search is positively correlated with precursor resilience ($r = .269$, $p < .001$), improvisation resilience ($r = .225$, $p < .01$), radical green innovation ($r = .278$, $p < .001$), and incremental green innovation ($r = .207$, $p < .01$). Precursor resilience is positively correlated with radical green innovation ($r = .396$, $p < .001$) and incremental green innovation ($r = .355$, $p < .001$). Moreover, improvisation resilience is positively correlated with radical green innovation ($r = .335$, $p < .001$) and incremental green innovation ($r = .335$, $p < .001$).

4.2 | Hypothesis test

We conducted hierarchical regression to examine the hypotheses and followed the suggestion of Haans et al. (2016) to test the inverted U-shaped relationship. The curvilinear relationship is expressed as $Y = \beta_0 + \beta_1 X + \beta_2 X^2$. According to Haans et al. (2016), three conditions should be satisfied to confirm the inverted U shape. First, the coefficient β_2 should be significant and negative. Second, the slope $\beta_1 + 2\beta_2 X_L$ should be significant and positive, and the slope $\beta_1 + 2\beta_2 X_H$ should be significant and negative, where X_L and X_H are the minimum and maximum values of X. Third, the turning point, $-\beta_1/2\beta_2$, should be within the range of X.

TABLE 2 Reliability and construct validity analysis

Variables	Items	Factor loadings	Cronbach's alpha	CR	AVE
Proactive boundary-spanning search (PBS)	PBS1	0.797	0.849	0.885	0.657
	PBS2	0.849			
	PBS3	0.824			
	PBS4	0.771			
Precursor resilience (PR)	PR1	0.809	0.938	0.920	0.725
	PR2	0.856			
	PR3	0.873			
	PR4	0.877			
	PR5	0.841			
Improvisation resilience (IR)	IR1	0.820	0.951	0.944	0.773
	IR2	0.895			
	IR3	0.898			
	IR4	0.905			
	IR5	0.875			
Radical green innovation (RGI)	RGI1	0.829	0.877	0.891	0.672
	RGI2	0.805			
	RGI3	0.868			
	RGI4	0.775			
Incremental green innovation (IGI)	IGI1	0.825	0.903	0.914	0.727
	IGI2	0.862			
	IGI3	0.888			
	IGI4	0.834			

TABLE 3 Discriminant validity analysis

Model	χ^2	*df*	RMSEA	TLI	CFI
One-factor model (PBS + PR + IR + RGI + IGI)	2602.477	209	0.230	0.313	0.378
Three-factor model (PBS, PR + IR, RGI + IGI)	1889.025	206	0.194	0.510	0.563
Four-factor model (PBS, PR + IR, RGI, IGI)	1364.870	203	0.162	0.657	0.698
Four-factor model (PBS, PR, IR, RGI + IGI)	923.407	203	0.128	0.787	0.813
Five-factor model (PBS, PR, IR, RGI, IGI)	473.597	199	0.080	0.917	0.929

TABLE 4 Means, standard deviations, and correlations

Variables	1	2	3	4	5	6	7	8	9	10
1. Firm age	1									
2. Firm scale	0.279***	1								
3. State-owned firms	0.147*	−0.044	1							
4. Private firms	−0.083	−0.010	−0.290***	1						
5. Sino-foreign firms	0.022	−0.140*	−0.357***	−0.376***	1					
6. PBS	0.006	0.216**	−0.001	−0.003	−.121	**0.811**				
7. PR	0.042	0.044	−0.105	0.035	0.072	0.269***	**0.851**			
8. IR	0.201**	0.136*	−0.027	0.022	0.120	0.225**	0.379***	**0.879**		
9. RGI	0.018	−0.042	−0.010	−0.034	0.139*	0.278***	0.396***	0.335***	**0.820**	
10. IGI	0.120	0.141*	−0.085	0.054	0.178**	0.207**	0.355***	0.335***	0.242***	**0.853**
Mean	3.710	3.110	0.216	0.234	0.317	3.900	3.583	3.483	3.565	3.810
S.D.	1.335	1.507	0.412	0.424	0.466	0.603	0.765	0.864	0.656	0.700

Note: The diagonal numbers in bold are the square root of AVE.
$^*p < .05$. $^{**}p < .01$. $^{***}p < .001$.

TABLE 5 Regression results

	PR			IR			RGI					IGI				
Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Mode 9	Model 10	Model 11	Model 12	Model 13	Model 14	Model 15	Model 16
Control Variables																
Firm age	0.044	0.070	0.056	0.163*	0.181*	0.168*	0.013	0.040	0.024	−0.037	−0.021	0.071	0.085	0.069	0.030	0.030
Firm scale	0.038	−0.042	−0.029	0.125	0.069	0.082	−0.017	−0.100	−0.085	−0.056	−0.092	0.169*	0.126	0.141*	0.137*	0.135*
State-owned firms	−0.073	−0.120	−0.108	0.069	0.037	0.048	0.068	0.020	0.033	0.076	0.052	0.077	0.053	0.065	0.085	0.083
Private firms	0.043	0.003	0.024	0.136	0.109	0.129	0.054	0.013	0.036	0.011	0.006	0.197*	0.176*	0.199*	0.162*	0.173*
Sino-foreign firms	0.067	−0.010	−0.011	0.210*	0.157	0.156	0.181*	0.102	0.100	0.114	0.075	0.302**	0.261**	0.260**	0.247**	0.238**
Independent variables																
PBS		0.279***	0.265***		0.190**	0.178**		0.288***	0.273***		0.172**		0.148*	0.134*		0.044
PBS^2			−0.188**			−0.177**			−0.207**		−0.125*			−0.204**		−0.132*
Mediator variables																
PR										0.316***	0.259***				0.265***	0.235**
IR										0.218**	0.183**				0.179**	0.155*
R^2	0.018	0.088	0.122	0.076	0.109	0.140	0.024	0.098	0.140	0.215	0.254	0.090	0.110	0.151	0.222	0.240
R^2 **change**		0.070***	0.034**		0.033**	0.031**		0.074***	0.042**	0.191***	0.114***		0.020*	0.041**	0.132***	0.089***
F	0.760	3.388**	4.185***	3.503**	4.308***	4.878***	1.027	3.842**	4.889***	8.208***	7.862***	4.215**	4.361***	5.335***	8.581***	7.306***

*$p < .05$. **$p < .01$. ***$p < .001$.

H1a and H1b indicate that proactive boundary-spanning search has an inverted U-shaped relationship with both radical and incremental green innovations. As displayed in Table 5, Model 9 and Model 14 indicate that proactive boundary-spanning search squared negatively affects radical green innovation ($\beta = -.207$, $p < .01$) and incremental green innovation ($\beta = -.204$, $p < .01$). The value of proactive boundary-spanning search ranges from −3.563 to 1.823 after mean-centered. When the values of proactive boundary-spanning search are the minimum and maximum, for radical green innovation, the slopes are 1.011 at the minimum −3.563 and −0.104 at the maximum 1.823, respectively, and the turning point is 0.659, within the range of proactive boundary-spanning search. With regard to incremental green innovation, the slopes are 0.861 at the minimum −3.563 and −0.238 at the maximum 1.823, respectively, and the turning point is 0.328, which is in the range of proactive boundary-spanning search. Therefore, H1a and H1b are supported. Further, Model 3 and Model 6 show that proactive boundary-spanning search squared negatively affects precursor resilience ($\beta = -.188$, $p < .01$) and improvisation resilience ($\beta = -.177$, $p < .01$). For precursor resilience, the slopes are 0.935 at the minimum −3.563 and −0.078 at the maximum 1.823, respectively, and the turning point 0.705 is within the range of proactive boundary-spanning search. Similarly, for improvisation resilience, the slopes are 0.809 at the minimum −3.563 and −0.145 at the maximum 1.823, respectively, and the turning point is 0.503, which is in the range of proactive boundary-spanning search. Therefore, H2a and H2b are supported.

To test the mediating role of organizational resilience, we conducted the procedure suggested by Baron and Kenny (1986). As indicated in Table 5, Model 9 shows that proactive boundary-spanning search squared has negative impact on radical green innovation ($\beta = -.207$, $p < .01$); Model 11 shows that both precursor resilience and improvisation resilience have positive effects on radical green innovation ($\beta = .259$, $p < .001$; $\beta = .183$, $p < .01$), and the negative impact of proactive boundary-spanning search squared on radical green innovation decreases but still significant ($\beta = -.125$, $p < .05$), proving that both precursor resilience and improvisation resilience play partial mediating role in proactive boundary-spanning search and radical green innovation. Thus, H3a and H3c are proved. Similarly, Model 14 shows that proactive boundary-spanning search squared has negative impact on incremental green innovation ($\beta = -.204$, $p < .01$); Model 16 shows that both precursor resilience and improvisation resilience have positive effects on incremental green innovation ($\beta = .235$, $p < 0.01$; $\beta = .155$, $p < .05$), and the negative impact of proactive boundary-spanning search squared on incremental green innovation decreases but still significant ($\beta = -.132$, $p < .05$), which suggests that precursor resilience and improvisation resilience play partial mediating role in proactive boundary-spanning search and incremental green innovation; thus, H3b and H3d are verified.

Hayes and Preacher (2010) believed that the method of testing mediating effect proposed by Baron and Kenny (1986) may distort the inverted U-shaped relationship and cannot clearly reveal the mediating effect between independent and dependent variables. Hence, we used PROCESS of SPSS 23.0 and conducted bootstrap analysis with 5000 replications and 95% confidence interval (CI) to further test the hypotheses. As displayed in Table 6, proactive boundary-spanning search squared has a significant negative impact on radical green innovation ($\beta = -.060$, $p < .05$) and incremental green innovation ($\beta = -.067$, $p < .05$), with 95% CI of [−.119, −.001] and [−.130, −.004], respectively. Each CI does not exclude zero, which indicates that proactive boundary-spanning search has an inverted U-shaped effect on radical green innovation and incremental green innovation, thus further supporting H1a and H1b. Moreover, proactive boundary-spanning search squared has a significant negative impact on precursor resilience ($\beta = -.114$, $p < .01$) and improvisation resilience ($\beta = -.119$, $p < .01$), with 95% CI of [−.188, −.041] and [−.120, −.038], respectively. Each CI does not exclude zero, which suggests that proactive boundary-spanning search has an inverted U-shaped impact on precursor resilience and improvisation resilience. Thus, H2a and H2b are further proved.

Regarding the mediating role of organizational resilience, as shown in Table 6, boundary-spanning search squared has an indirect effect on radical green innovation via precursor resilience ($\beta = -.029$) and improvisation resilience ($\beta = -.018$), with 95% CI of [−.065, −.008] and [−.053, −.001], respectively, and the total effect is −0.107 with 95% CI of [−.170, −.044]. Each CI does not contain zero, supporting H3a and H3c. Moreover, the results indicate that boundary-spanning search squared has an indirect effect on incremental green innovation via precursor resilience ($\beta = -.026$) and improvisation resilience ($\beta = -.015$), with 95% CI of [−.064, −.004] and [−.041, −.001], respectively, and the total effect is −0.108 with 95% CI of [−.173, −.043]. Each CI does not contain zero; hence, H3b and H3d are further verified.

5 | DISCUSSION

5.1 | Theoretical implications

First, drawing on RBV, this study explores the antecedents of green innovation by emphasizing proactive boundary-spanning search. Unlike previous studies (Wang et al., 2020; Yang et al., 2021), we explore the influencing factors of green innovation from resources and proactive behavior perspectives. The results reveal that proactive boundary-spanning search is a critical driver of radical and incremental green innovations and exerts inverted U-shaped effect on both radical and incremental green innovations. Thus, this finding expands the research of green innovation by uncovering the nonlinear effects of proactive boundary-spanning search on two types of green innovation.

Second, we distinguish two types of organizational resilience, namely, precursor resilience and improvisation resilience, to deeply understand organizational resilience. Furthermore, we find that proactive boundary-spanning search is not always beneficial to precursor resilience and improvisation resilience, and there is inverted U-shaped impact of proactive boundary-spanning search on precursor resilience and improvisation resilience. In light of this, both precursor resilience and improvisation resilience can be enhanced by moderate proactive

TABLE 6 Bootstrap results of mediation effects

Path	Direct effect	SE	t	LLCI	ULCI
$PBS^2 \rightarrow$ RGI	−0.060	0.030	−1.990*	−0.119	−0.001
$PBS^2 \rightarrow$ IGI	−0.067	0.032	−2.107*	−0.130	−0.004
$PBS^2 \rightarrow$ PR	−0.114	0.038	−3.049**	−0.188	−0.041
$PBS^2 \rightarrow$ IR	−0.119	0.041	−2.886**	−0.120	−0.038
	Indirect effect	**Boot SE**		**Boot LLCI**	**Boot ULCI**
$PBS^2 \rightarrow$ PR $\rightarrow$ RGI	−0.029	0.014		−0.065	−0.008
$PBS^2 \rightarrow$ IR $\rightarrow$ RGI	−0.018	0.013		−0.053	−0.001
$PBS^2 \rightarrow$ PR $\rightarrow$ IGI	−0.026	0.015		−0.064	−0.004
$PBS^2 \rightarrow$ IR $\rightarrow$ IGI	−0.015	0.010		−0.041	−0.001
	Total effect	**SE**	**t**	**LLCI**	**ULCI**
$PBS^2 \rightarrow$ RGI	−0.107	0.032	−3.353***	−0.170	−0.044
$PBS^2 \rightarrow$ IGI	−0.108	0.033	−3.292**	−0.173	−0.043

Note: SE, Standard error; LLCI, Lower limit CI; ULCI, Upper limit CI.
$^*p < .05$. $^{**}p < .01$. $^{***}p < 0.001$.

boundary-spanning search. Our research therefore enriches organizational resilience literature by emphasizing the importance of proactive boundary-spanning search and confirms that proactive boundary-spanning search is a vital antecedent of organizational resilience.

Third, our research uncovers the "black box" of proactive boundary-spanning search on firms' green innovation by highlighting organizational resilience. We not only analyze the direct impact of proactive boundary-spanning search on firms' green innovation, but also emphasize organizational resilience and clarify its mediating role in proactive boundary-spanning search and firms' green innovation. Previous studies have explored the direct impact of external knowledge on green innovation (Ben Arfi et al., 2018; Wang et al., 2020), but ignored the role of organizational resilience in the VUCA environment. Drawing on RBV and dynamic capabilities theory, we regard proactive boundary-spanning search as a way to obtain resources and organizational resilience as a dynamic capability, and explain how resources acquisition and dynamic capability enhance green innovation. Our study expands the existing research on green innovation by considering both proactive boundary-spanning search and organizational resilience and provides novel insights for green innovation.

5.2 | Managerial implications

First, we find that proactive boundary-spanning search has inverted U-shaped impact on radical green innovation and incremental green innovation. This implies that proactive boundary-spanning search is particularly important for green innovation and can bring rich knowledge resources for firms to conduct green innovation. However, too much proactive boundary-spanning search does not lead to the best green innovation. In fact, moderate proactive boundary-spanning search is most conducive to firms' green innovation. Specifically, when proactive boundary-spanning search is moderate, firms can obtain the required heterogeneous knowledge and use it to the greatest extent, improving both radical green innovation and incremental green innovation. Therefore, in order to facilitate green innovation, on the one hand, managers should regard knowledge acquisition as their primary activities and search new knowledge ahead of their competitors; on the other hand, managers should pay attention to the level of proactive search activities and keep them within an appropriate range.

Second, we find that there is an inverted U-shaped relationship between proactive boundary-spanning search and organizational resilience, and organizational resilience plays a mediating role in boundary-spanning search and green innovation. If proactive boundary-spanning search can be controlled at an appropriate level, organizational resilience will positively affect radical and incremental green innovations. Therefore, organizational resilience, as a dynamic capability, enables firms to handle unexpected events when firms conduct green innovation under highly turbulence environment. Thus, to survive in the VUCA environment, especially in the recent COVID-19 pandemic crisis, managers should not overlook organizational resilience and should foster precursor resilience and improvisation resilience simultaneously. Specifically, on the one hand, managers should anticipate unpredictable events and boost preparedness, such as developing detailed emergency plans, monitoring early signals of emergencies, and detecting abnormal events. On the other hand, managers should cultivate adaptive capability and improvisation capability to enhance responsiveness, as these capabilities are critical during unexpected events. In this way, firms can effectively handle unexpected events and realize sustainable development.

5.3 | Limitations and future research

Our research has some limitations. First, we select manufacturing industry as the research context, and hence, the results may not be generalized to other industries. In the future, the relationship between proactive boundary-spanning search and green innovation in other industries should be studied. Second, we only consider the mediating role of organizational resilience; other factors, such as green dynamic capability and resource orchestration capability, can be taken into account in the future. Finally, to better understand the relationship between proactive boundary-spanning search and green innovation, it is worth collecting longitudinal data.

6 | CONCLUSIONS

This study empirically analyzes the relationship between proactive boundary-spanning search and green innovation and investigates the mediating role of organizational resilience. First, our results highlight that proactive boundary-spanning search has an inverted U-shaped effect on both radical and incremental green innovations. This is consistent with Yang et al. (2021), who suggested that boundary-spanning search can facilitate green innovation in an inverted U-shaped way. Our results are also partially consistent with Cainelli et al. (2015) and Liao (2018), who argued that external knowledge spurs green innovation. Our findings further confirm that there is a curvilinear relationship between proactive boundary-spanning search and green innovation, rather than a linear relationship. Second, we prove that proactive boundary-spanning search has inverted U-shaped effect on precursor resilience and improvisation resilience. This means that only a moderate level of proactive boundary-spanning search can facilitate organizational resilience, which is consistent with the results of Duchek (2020) and Jia et al. (2020), who proposed that knowledge contributes to organizational resilience. We also emphasize the importance of external knowledge and reveal that proactive boundary-spanning search is practically important for firms to build resilience. Third, our research demonstrates that both precursor resilience and improvisation resilience play partial mediating role in proactive boundary-spanning search, radical green innovation and incremental green innovation. This aligns with the research of Do et al. (2022), who argued that organizational learning affects organizational resilience, and organizational resilience further promotes innovation. Our research echoes the suggestion that knowledge search not only has direct effect on green innovation but also has indirect impact on green innovation. The findings provide evidence to better explain why some firms can use external knowledge for green innovation while others cannot.

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CONFLICT OF INTEREST

The authors declare that there are no conflicts of interest.

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